

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q20-1561

Bedford Park 100 kV
December 02, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1561. The Interconnection Customer proposes a 95.7 MW wind facility interconnecting at the Bedford Park 100 kV substation in the Illinois area. The request was received on August 30, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 539 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$110,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Royse to Athol 100kV L	Opal Cliffs to Morganton Parker Rd 100kV	108.8	141	8.1
Opal Cliffs to Morganton Parker Rd 100kV	Hayford to Ceres 100kV	107.5	131	19.6

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Royse to Athol 100kV L	Reconductor line	\$391,000	11% (\$45,000)
Opal Cliffs to Morganton Parker Rd 100kV	Reconductor line	\$267,000	24% (\$65,000)
	Total allocated to project		\$110,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.